P2 Chapter 9: Differentiation

The Chain Rule

The Chain Rule

The chain rule allows us to differentiate a composite function, i.e. a function within a function.

$$y = (3x^4 + x)^5$$

The Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Fro Note: Notice how the du's sort of 'cancel' top and bottom on the RHS. This is <u>not</u> a valid proof of the chain rule, but the d's sort of behave as quantities which can often be manipulated in this way.

Full Method:

Let $u = 3x^4 + x$

In the Chain Rule, u represents the inner function.

$$\frac{du}{dx} = 12x^3 + 1$$

$y = u^5$

This represents the 'outer' function.

$$\frac{dy}{du} = 5u^4 = 5(3x^4 + x)^4$$

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= 5(3x^4 + x)^4(12x^3 + 1)$$

Doing it mentally in one go:

(aka the 'bla method')

The chain rule boils down to:

$\frac{dy}{dx}$ = outer function differentiated

$\times$ inner function differentiated

$$\frac{dy}{dx} = 5(3x^4 + x)^4 \times (12x^3 + 1)$$

When differentiating the outer function, replace (in your head) the inner function with the word 'bla'. i.e. "What does bla^5 differentiate to?". It's $5\ bla^4$

Now differentiate the inner function, i.e. the 'bla'.

Further Practice

	Outer Function Differentiated		Inner Function Differentiated
$y = (x^2 + 1)^3$	$\frac{dy}{dx} =$?	$\times$	?

The outer function is "bla^3"

$y = (\ln x)^3$	$\frac{dy}{dx} =$?	$\times$	?

The outer function is again "bla^3"

$y = e^{x^2+x}$	$\frac{dy}{dx} =$?	$\times$	?

The outer function is "e^{bla}"

$y = (2^x + 1)^2$	$\frac{dy}{dx} =$?	$\times$	?
$y = \ln(\sin x)$	$\frac{dy}{dx} =$?	$\times$	?

Further Practice

	Outer Function Differentiated	Inner Function Differentiated
$y = (x^2+1)^3$	$\frac{dy}{dx} = 3(x^2+1)^2$	$\times\ 2x$
$y = (\ln x)^3$	$\frac{dy}{dx} = 3(\ln x)^2$	$\times\ \frac{1}{x}$
$y = e^{x^2+x}$	$\frac{dy}{dx} = e^{x^2+x}$	$\times\ (2x+1)$
$y = (2^x+1)^2$	$\frac{dy}{dx} = 2(2^x+1)$	$\times\ 2^x \ln 2$
$y = \ln(\sin x)$	$\frac{dy}{dx} = \frac{1}{\sin x}$	$\times\ \cos x$

The outer function is "bla^3"

The outer function is again "bla^3"

The outer function is "e^{bla}"

Yet Further Practice

	Outer Function Differentiated		Inner Function Differentiated
$y = \sin 5x$	$\frac{dy}{dx} =$?	$\times$	?

The outer function is "$\sin bla$"

This is consistent with what we learned earlier in the chapter.

$y = \sin^2 x =$?

$\frac{dy}{dx} =$? $\times$?

Tip: When differentiating a trig function to a power, always rewrite using a bracket first.

$y = \sqrt{x+1} =$?

$\frac{dy}{dx} =$? $\times$?

$y = \cos^3 2x =$?

$\frac{dy}{dx} =$? $\times$?

Here we had to use the Chain Rule twice!

$y = e^{e^x}$

$\frac{dy}{dx} =$? $\times$?

Yet Further Practice

Outer Function Differentiated

Inner Function Differentiated

$$y = \sin 5x \qquad \frac{dy}{dx} = \cos 5x \times 5$$

The outer function is "$\sin bla$"

This is consistent with what we learned earlier in the chapter.

$$y = \sin^2 x = (\sin x)^2 \qquad \frac{dy}{dx} = 2\sin x \times \cos x$$

Tip: When differentiating a trig function to a power, always rewrite using a bracket first.

$$y = \sqrt{x+1} = (x+1)^{\frac{1}{2}} \qquad \frac{dy}{dx} = \frac{1}{2}(x+1)^{-\frac{1}{2}} \times 1$$

Here we had to use the Chain Rule twice!

$$y = \cos^3 2x = (\cos 2x)^3 \qquad \frac{dy}{dx} = 3(\cos 2x)^2 \times (-2\sin 2x)$$

$$y = e^{e^x} \qquad \frac{dy}{dx} = e^{e^x} \times e^x$$

Test Your Understanding

C3 June 2011 Q1a

Differentiate with respect to x

(*a*) $\ln(x^2 + 3x + 5)$, **(2)**

?

In general:

$$\frac{d}{dx}(\ln(f(x)) = \frac{f'(x)}{f(x)}$$

[Textbook] Given that $y = \sqrt{5x^2 + 1}$, find $\frac{dy}{dx}$ at $(4,9)$

?

Test Your Understanding

C3 June 2011 Q1a

Differentiate with respect to x

(a) $\ln(x^2 + 3x + 5)$, **(2)**

In general:

$$\frac{d}{dx}(\ln(f(x)) = \frac{f'(x)}{f(x)}$$

$$\frac{dy}{dx} = \frac{1}{x^2 + 3x + 5} \times (2x + 3) = \frac{2x + 3}{x^2 + 3x + 5}$$

[Textbook] Given that $y = \sqrt{5x^2 + 1}$, find $\frac{dy}{dx}$ at $(4,9)$

$$y = (5x^2 + 1)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(5x^2 + 1)^{-\frac{1}{2}} \times 10x$$

When $x = 4$,

$$\frac{dy}{dx} = \frac{1}{2}(5(4^2) + 1)^{-\frac{1}{2}} \times 10(4) = \frac{20}{9}$$

dx/dy

$$\frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)}$$

Proof: By the chain rule:

$$\frac{dy}{dy} = \frac{dy}{dx} \times \frac{dx}{dy}$$

But $\frac{dy}{dy} = \frac{d}{dy}(y) = 1. \therefore \frac{dy}{dx} = 1 \div \frac{dx}{dy}$

Sometimes we might have x in terms of y, but we want to find $\frac{dy}{dx}$.

Find $\frac{dy}{dx}$ when $x = 2y^2 + y$

?

Find the gradient of $x = (1 + 2y)^3$ when $y = 1$

?

dx/dy

$$\frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)}$$

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Sometimes we might have x in terms of y, but we want to find $\frac{dy}{dx}$.

Find $\frac{dy}{dx}$ when $x = 2y^2 + y$

$$\boldsymbol{\frac{dx}{dy} = 4y + 1}$$

$$\boldsymbol{\frac{dy}{dx} = \frac{1}{4y + 1}}$$

Find the gradient of $x = (1 + 2y)^3$ when $y = 1$

$$\boldsymbol{\frac{dx}{dy} = 6(1 + 2y)^2} \qquad \rightarrow \qquad \boldsymbol{\frac{dy}{dx} = \frac{1}{6(1 + 2y)^2}}$$

$$\boldsymbol{m = \frac{1}{6(1 + 2)^2} = \frac{1}{54}}$$

Exercise 9.3

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Homework Exercise

1 Differentiate:

a $(1 + 2x)^4$ **b** $(3 - 2x^2)^{-5}$ **c** $(3 + 4x)^{\frac{1}{2}}$ **d** $(6x + x^2)^7$

e $\dfrac{1}{3 + 2x}$ **f** $\sqrt{7 - x}$ **g** $4(2 + 8x)^4$ **h** $3(8 - x)^{-6}$

2 Differentiate:

a $e^{\cos x}$ **b** $\cos(2x - 1)$ **c** $\sqrt{\ln x}$ **d** $(\sin x + \cos x)^5$

e $\sin(3x^2 - 2x + 1)$ **f** $\ln(\sin x)$ **g** $2e^{\cos 4x}$ **h** $\cos(e^{2x} + 3)$

3 Given that $y = \dfrac{1}{(4x + 1)^2}$ find the value of $\dfrac{dy}{dx}$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$.

4 A curve C has equation $y = (5 - 2x)^3$. Find the tangent to the curve at the point P with x-coordinate 1. **(7 marks)**

5 Given that $y = (1 + \ln 4x)^{\frac{3}{2}}$, find the value of $\dfrac{dy}{dx}$ at $x = \dfrac{1}{4}e^3$. **(5 marks)**

6 Find $\dfrac{dy}{dx}$ for the following curves, giving your answers in terms of y.

a $x = y^2 + y$ **b** $x = e^y + 4y$ **c** $x = \sin 2y$ **d** $4x = \ln y + y^3$

Homework Exercise

7 Find the value of $\frac{dy}{dx}$ at the point (8, 2) on the curve with equation $3y^2 - 2y = x$.

Problem-solving

Your expression for $\frac{dy}{dx}$ will be in terms of y.

Remember to substitute the y-coordinate into the expression to find the gradient.

8 Find the value of $\frac{dy}{dx}$ at the point $\left(\frac{5}{2}, 4\right)$ on the curve with equation $y^{\frac{1}{2}} + y^{-\frac{1}{2}} = x$.

9 **a** Differentiate $e^y = x$ with respect to y.

b Hence, prove that if $y = \ln x$, then $\frac{dy}{dx} = \frac{1}{x}$

10 The curve C has equation $x = 4\cos 2y$.

a Show that the point $Q\left(2, \frac{\pi}{6}\right)$ lies on C. **(1 mark)**

b Show that $\frac{dy}{dx} = -\frac{1}{4\sqrt{3}}$ at Q. **(4 marks)**

c Find an equation of the normal to C at Q. Give your answer in the form $ax + by + c = 0$, where a, b and c are exact constants. **(4 marks)**

Homework Exercise

11 Differentiate:

a $\sin^2 3x$ **b** $e^{(x+1)^2}$ **c** $\ln(\cos x)^2$

d $\dfrac{1}{3+\cos 2x}$ **e** $\sin\left(\dfrac{1}{x}\right)$

12 The curve C has equation $y = \dfrac{4}{(2-4x)^2}$, $x \neq \dfrac{1}{2}$

The point A on C has x-coordinate 3.
Find an equation of the normal to C at A in the form $ax + by + c = 0$, where a, b and c are integers. **(7 marks)**

13 Find the exact value of the gradient of the curve with equation $y = 3^{x^3}$ at the point with coordinates (1, 3). **(4 marks)**

Challenge

Find $\dfrac{dy}{dx}$ given that:

a $y = \sqrt{\sin\sqrt{x}}$ **b** $\ln y = \sin^3(3x+4)$

Homework Answers

1 **a** $8(1+2x)^3$ **b** $20x(3-2x^2)^{-6}$

c $2(3+4x)^{-\frac{1}{2}}$ **d** $7(6+2x)(6x+x^2)^6$

e $-\dfrac{2}{(3+2x)^2}$ **f** $-\dfrac{1}{2\sqrt{7-x}}$

g $128(2+8x)^3$ **h** $18(8-x)^{-7}$

2 **a** $-\sin x\, e^{\cos x}$ **b** $-2\sin(2x-1)$ **c** $\dfrac{1}{2x\sqrt{\ln x}}$

d $5(\cos x - \sin x)(\sin x + \cos x)^4$

e $(6x-2)\cos(3x^2-2x+1)$

f $\cot x$ **g** $-8\sin 4x\, e^{\cos 4x}$ **h** $-2e^{2x}\sin(e^{2x}+3)$

3 -1 **4** $y = -54x + 81$ **5** $12e^{-3}$

6 **a** $\dfrac{1}{2y+1}$ **b** $\dfrac{1}{e^y+4}$ **c** $\dfrac{1}{2}\sec 2y$ **d** $\dfrac{4y}{1+3y^3}$

7 $\frac{1}{10}$ **8** $\frac{16}{3}$

9 **a** $e^y = \dfrac{dx}{dy}$

b $y = \ln x$, $e^y = x$

Differentiate with respect to y using part **a**

$e^y = \dfrac{dx}{dy} \Rightarrow \dfrac{1}{e^y} = \dfrac{dy}{dx}$

Since $x = e^y$, $\dfrac{dy}{dx} = \dfrac{1}{x}$

10 **a** $4\cos 2\left(\dfrac{\pi}{6}\right) = 4\left(\dfrac{1}{2}\right) = 2$

When $y = \dfrac{\pi}{6}$, $x = 2$, therefore $\left(2, \dfrac{\pi}{6}\right)$ lies on C.

b $\dfrac{dx}{dy} = -8\sin 2y$

At $Q\left(2, \dfrac{\pi}{6}\right)$: $\dfrac{dx}{dy} = -8\sin 2\left(\dfrac{\pi}{6}\right) = -8\left(\dfrac{\sqrt{3}}{2}\right) = -4\sqrt{3}$

So, $\dfrac{dy}{dx} = -\dfrac{1}{4\sqrt{3}}$

c $4\sqrt{3}\,x - y - 8\sqrt{3} + \dfrac{\pi}{6} = 0$

11 **a** $6\sin 3x\cos 3x$ **b** $2(x+1)e^{(x+1)^2}$ **c** $-2\tan x$

d $\dfrac{2\sin 2x}{(3+\cos 2x)^2}$ **e** $-\dfrac{1}{x^2}\cos\left(\dfrac{1}{x}\right)$

12 $3125x - 100y - 9371 = 0$ **13** $9\ln 3$

Challenge

a $\dfrac{\cos\sqrt{x}}{4\sqrt{x}\sin\sqrt{x}}$ **b** $9e^{\sin^3(3x+4)}\cos(3x+4)\sin^2(3x+4)$